

3.8 Topic 6: The Technological Fix?

Technology and development

Humans increasingly depend on new technology, and access to technology is closely related to level of development. Just as development is distinctly uneven, so is the geography of technology. Many people have the expectation that technology will help them, and solve problems, whereas others lack access to technological innovation at even basic levels. The question of who should have access to technology, and at what price, is a key one.

Use of technology

Use of technology has costs as well as benefits, both social and environmental. These may be known, but in some cases are unforeseen. Technology can be used in a variety of ways, an obvious contrast is between large-scale top-down mega-projects and small scale intermediate and more appropriate approaches. Technology may have the potential to solve some key geographical problems and issues, but is unlikely to be accessible to all, and the desirability of large-scale technological fixes needs to be assessed.

1 The geography of technology

Enquiry question: *Why is there inequality in access to technology?*

What students need to learn	Suggested teaching and learning
■ Technology can be defined as the ability to control nature; technology is pervasive, as is dependency on modern technology; this varies significantly between countries and regions.	■ Defining technology, and developing an understanding of its nature and geographical spread.
■ Geographical distribution of technology use at a variety of scales (national and global contrasts through farming, telecommunications and transport connectivity) shows distinct patterns related to level of development.	■ Investigating those who are connected and disconnected in terms of a range of technologies.
■ Access to technology varies in different parts of the world and there is a contrast between those who can access new technology versus those who are left to suffer from environmental determinism.	■ Developing an appreciation of the consequences of access to technology, resulting in differences in lifestyle, wealth and security.

■ There is a range of reasons for inequality of access such as cost barriers, intellectual property and the patent system; knowledge, education, political or religious denial of access. Some players may have little interest in 'sharing' technological innovation, despite its benefits.	■ Examining and illustrating the debate over access to technology; the role of governments and other players including moral aspects, for example Aids/HIV treatment, nuclear technology, GM crops and the internet.
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2 Technology and development

Enquiry question: *How far does technology determine development and resource use?*

What students need to learn	Suggested teaching and learning
■ There is a link between economic development and technological innovation, and evidence of a widening technology gap in increasingly knowledge-based economies.	■ Exploring the link between technology and development, for instance via patent grants, government sponsorship of research and development and participation rates in higher education, to show that technology is one of the important drivers of development.
■ Technological leapfrogging may be a way of overcoming some of the barriers to development.	■ Investigating how the technology gap might be reduced by leapfrogging for example mobile phones in Asia; GM crop use in Latin America and Africa.
■ Technological innovation may have unforeseen social, environmental and economic costs and benefits.	■ Developing an understanding, using examples, that much technology is not 'neutral' for example the green revolution and GM crop technology.
■ The externalities of technology use are accounted for in some economies but not in others, with contrasting consequences for human and ecosystem wellbeing.	■ Investigating different approaches to externalities and their impacts, such as 'polluter pays' versus treating the environment as a sink.

3 Technology, environment and the future

Enquiry question: *What is the role of technology in the management of the contested planet?*

What students need to learn	Suggested teaching and learning
■ There are contrasts between appropriate/ intermediate technology approaches versus megaprojects as models for development, in terms of environmental impacts and social equity.	■ Weighing up the role of basic technology and comparing this to high-tech/top-down development (dams, water transfer schemes) in terms of social and environmental consequences.
■ Technology may have a direct role in overcoming global environmental issues, such as global warming and land degradation – but technological fixes may not be feasible or desirable.	■ Examining technological fixes proposed to solve global problems and setting these against criteria such as those for environmental sustainability.
■ The question of whether increasing technology use and an environmentally sustainable future can be reconciled — or are mutually exclusive — is an important one.	■ Examining whether technology will always lead to waste and pollution, or if it can be developed within a more sustainable model.
■ The future may be a divergent one (core 'technologically fixed' world versus a peripheral 'technologically impoverished' world); or a convergent one (technology for all).	■ Exploring a range of technological futures ranging from 'business as usual' divergence to global technological convergence.